

$$v_i = x_i + y_i$$
$$h_i \in \{2, 1\}$$

$$v_j = x_j + y_j$$
$$h_j \in \{2, 1\}$$

$$e_{ij} = v_i + v_j$$
$$r_{ij} = h_i + h_j \leq 3$$

Canonical output
one edge normalization